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Máire Ní Ríordáin

Craig Neville

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An operational framework for integrating content, cognition and language into planning for AEN mathematics learners within a CLIL context

Máire Ní Ríordáin*; Craig Neville

University College Cork

Abstract: The growing diversity in Content and Language Integrated Learning (CLIL) classrooms, particularly the inclusion of Additional Educational Needs (AEN) learners, highlights an urgent need for research addressing their unique challenges. These students often face compounded difficulties, navigating the dual demands of mastering subject content and developing proficiency in the instructional language, with limited frameworks to support their needs. This article presents an operational framework for integrating content, cognition, and language into planning mathematics education for AEN learners. Grounded in theoretical approaches such as communicative competence, CAPA, and pluriliteracies, the framework addresses key challenges in developing mathematical literacy. The framework provides a structured approach for planning in multilingual classrooms, offering a foundation for educators and policymakers to create equitable and effective learning environments that support mathematics learners.

Keywords: CLIL; mathematical literacy; AEN learners; inclusive classrooms

Introduction

Learning mathematics through the medium of an additional language is a widespread practice globally, however occurring for diverse purposes. In Europe, formal bilingual approaches are often termed Content and Language Integrated Learning (CLIL) (Gerns, 2023), where academic subjects like mathematics are taught through a target language, typically English. Irish-medium education (IME) includes Gaeltacht schools (heritage language maintenance) and Gaelscoileanna (immersion education), where Irish is the instructional language. Gaelscoileanna are located in English-speaking communities, and students are generally from English-speaking backgrounds. Thus, IME aligns with CLIL's "dual-focused educational approach", combining content and language learning (Coyle, Hood & Marsh, 2010, p. 1). In IME, all subjects except English are taught through Irish.

* maire.niriordain@ucc.ie

CLIL has been perceived as catering predominantly to high-achieving students (Madrid & Pérez Cañado, 2018), but its student demographics have diversified (Pérez Cañado, 2023). For example, 9.4% of IME primary students are identified with AEN, rising to 16.6% when including those receiving extra support without formal diagnoses, comparable to English-medium schools (Nic Aindriú, Ó Duibhir & Travers, 2024). AEN students in CLIL settings face compounded challenges, making them a minority within a minority (Barnes, 2024). Research across European bilingual education highlights issues of inclusivity, lack of resources, and teacher preparedness to support diverse learners (Pérez Cañado, 2023). These findings suggest that a one-size-fits-all approach is inadequate, emphasizing the need for tailored strategies to better support AEN learners in CLIL mathematics contexts.

According to Planas and Abtahi (2024), research in mathematics education should be reframed to incorporate a wider range of learners from different linguistic and educational backgrounds to foster a more equitable, diverse, and inclusive field; we seek to address this partially here. This article describes an operational framework for integrating content, cognition, and language into planning for AEN mathematics learners in CLIL contexts. Drawing on theoretical approaches for language and communication planning, this work integrates mathematical literacy and builds upon findings from a research study with AEN mathematics learners in a Gaelscoil (Ní Riórdáin et al., 2024). While developed for CLIL, the framework may also benefit AEN multilingual mathematics learners in other settings.

Addressing inclusivity challenges for AEN learners in CLIL mathematics education

Despite CLIL's growing adoption across diverse educational settings, challenges related to inclusivity, particularly for students with AEN, persist (Pérez Cañado, 2023). These challenges raise significant concerns about the equity and effectiveness of CLIL in addressing the needs of all learners (Nic Aindriú & Ó Duibhir, 2023). Studies conducted across Europe have highlighted a noticeable shortfall in the availability of instructional materials, adaptable methodologies, and teacher-training programmes that address the varied learning profiles found in classrooms (Pérez Cañado, 2023). As CLIL expands beyond elite settings to include mainstream schools, its ability to support students with AEN remains limited, with few tailored interventions or assessments in place (Nic Aindriú & Ó Duibhir, 2023; Pérez Cañado, 2023).

In the specific context of IME, these issues are particularly pressing. The challenges for AEN learners in IME are outlined by Nic Aindriú and Ó Duibhir (2023), which are amplified by the scarcity of resources, such as bilingual assessments and subject-specific learning tools in Irish. They also report that teachers struggle to assess and support students effectively, particularly in key areas like literacy and mathematics. Moreover, professional development for teachers in IME settings has not kept pace with the increasing diversity of the student population. Teachers often lack the training to adapt CLIL methodologies to meet the dual demands of language and content learning for AEN students. This gap not only places additional strain on educators, but also risks leaving AEN learners without adequate support, limiting their access to the full benefits of immersion education (Pérez Cañado, 2023).

The challenges of integrating language and mathematics in CLIL classrooms are compounded by the persistence of traditional approaches that treat content knowledge and language proficiency as separate domains. In mathematics education, this has led to an overemphasis on computational skills and procedural fluency at the expense of conceptual understanding and critical reasoning (Lenz et al., 2024; Moschkovich, 2015). For AEN learners, this disjointed approach is particularly detrimental, as they are expected to navigate a cognitively demanding subject using a language in which they may lack proficiency. Existing research has shown that learners with limited language proficiency are disproportionately disadvantaged when confronted with the dual demands of abstract mathematical reasoning and academic language (Lenz et al., 2024). Additionally, Ingram et al. (2024) highlight that linguistic barriers in mathematics classrooms extend beyond multilingual learners to include students with limited access to academic mathematical discourse. While studies advocate for language-responsive instruction, these approaches often lack specificity in addressing the cognitive and representational demands of mathematics (Prediger & Wessel, 2013). This research underscores the urgency of developing planning strategies that integrate content, cognition, and language as mutually reinforcing components for AEN learners in CLIL contexts, and which may be of benefit in other settings where linguistic challenges exist.

The role of academic language in mathematics highlights another critical dimension of this issue. Mathematics has its own register, characterized by precise terminology, complex syntactic structures, and multimodal representations (Moschkovich, 2015). For AEN learners in CLIL classrooms, mastering this register is particularly challenging, as they must simultaneously develop basic linguistic competence in the instructional language. Existing interventions tend to

focus narrowly on vocabulary acquisition or syntactic patterns, neglecting the deeper, situated meanings that mathematical discourse entails. This reductionist view not only limits learners' ability to articulate their reasoning, but also undermines their capacity to engage with the conceptual underpinnings of mathematics (Lenz et al., 2024). Simplifying linguistic demands without diluting content allows learners to focus on conceptual understanding, a critical area often neglected in traditional mathematics instruction (Erath et al., 2021). A more integrated approach that emphasizes the cognitive and discursive dimensions of mathematical learning is therefore essential (Moschkovich, 2015).

Multimodal scaffolding is a promising but underutilized strategy for bridging linguistic and cognitive barriers in CLIL mathematics education. The use of visual aids, manipulatives, and symbolic representations can reduce linguistic complexity without lowering cognitive demand, thereby supporting learners in accessing abstract concepts (Lenz et al., 2024). However, current practices often fail to leverage these resources systematically. For example, visual aids are frequently used but are rarely integrated into a coherent instructional framework that aligns with learners' linguistic and cognitive needs. Differentiation strategies are also widely recommended, yet their implementation in mathematics education remains inconsistent (Lenz et al., 2024).

Tiered tasks, for instance, designed to support varying proficiency levels, often fail to address the intersectional challenges of language and cognitive demand, particularly for learners with additional needs. This results in a fragmented learning experience that neither advances conceptual understanding, nor supports language development. Furthermore, transitions between everyday language, the school register, and the technical mathematical register are not explicitly scaffolded, leaving many learners unable to make connections between their intuitive understandings and formal mathematical representations (Prediger & Wessel, 2013).

Current CLIL methodologies also frequently adopt a standardized model that fails to account for students' cultural and linguistic diversity (Moschkovich, 2021). This not only reduces instructional effectiveness, but also risks alienating learners by disregarding their prior knowledge and experiences. Planas and Abtahi (2024) argue that dominant research often overlooks language-responsive teaching strategies that accommodate diverse linguistic backgrounds. They advocate for decentralizing research to recognize the experiences of educators working in underrepresented contexts. Culturally and linguistically responsive planning, which incorporates students' home

languages and real-world contexts, can enhance engagement while supporting academic language development (Prediger & Wessel, 2013). However, such approaches require careful balancing to maintain both linguistic inclusivity and cognitive rigor.

Addressing the inclusivity challenges faced by AEN learners in CLIL, mathematics education requires a multifaceted approach. Language-responsive teaching, sociocultural perspectives, multimodal scaffolding, and culturally responsive planning all contribute to a more inclusive pedagogical framework. The persistence of traditional CLIL approaches, which often separate content and language learning, disproportionately disadvantages AEN students. Developing an operational framework for integrating content, cognition and language into planning for AEN mathematics learners within a CLIL context is essential. By advancing these principles, this research contributes to the ongoing discourse on inclusive education and the need for pedagogical frameworks that support AEN learners effectively.

The case study

A case study explored the impact of a language-responsive mathematics teaching intervention and language-adapted assessments on AEN learners in a Gaelscoil (Irish-medium education setting), focusing on mathematics taught through Irish as a second language (see Ní Ríordáin et al., 2024). Five students (aged approximately nine) with identified AEN, including dyslexia, ADHD, autism, and language processing difficulties, participated. In this study, one of the authors served as the teacher–researcher, collaborating closely with the class teacher to design and implement the intervention. The teacher–researcher’s role involved co-planning lesson activities, embedding language-responsive strategies into the students’ existing support plans, and conducting reflective discussions with the class teacher to refine instructional approaches. The class teacher played a critical role in ensuring that the intervention aligned with the students’ ongoing learning experiences. Together, the teacher–researcher and class teacher worked to adapt both instruction and assessment to the linguistic and cognitive needs of the students. This collaborative model ensured that the intervention was not an isolated program, but rather a contextually embedded approach within the students’ broader learning trajectory.

The five students participated in a small-group mathematics session where targeted instruction was provided, four times per week, over the course of five weeks (see Ní Ríordáin et al., 2024, for further detail). Outside of these sessions, they participated in the main classroom setting, engaging

in teacher- and peer-supported learning activities and whole-class instruction and discussion. Their distribution across these instructional formats was based on their mathematical proficiency levels, their language proficiency and specific linguistic needs, and the instructional focus of each lesson. This flexible grouping approach ensured that students received targeted support in the small-group sessions, while also benefiting from collaborative problem solving with their peers in the main classroom setting.

The intervention was structured to complement these pre-existing support plans, integrating language-responsive teaching strategies that explicitly scaffolded students' mathematical discourse to ensure that language adaptations were directly relevant to their cognitive and academic needs. These lessons incorporated structured sentence frames, visual supports, and explicit modelling of mathematical explanations to help students transition from informal to formal mathematical language. Students engaged in problem-solving activities that were designed to align with their linguistic and mathematical readiness, ensuring equitable access to mathematical discourse. Finally, assessment questions were modified to reduce unnecessary linguistic complexity while maintaining mathematical rigor. Students were encouraged to explain their reasoning using structured prompts and visual supports.

The case study revealed several insights into the challenges and opportunities for AEN learners in a CLIL mathematics classroom context. Language-adapted assessments, while enhancing question comprehension, had minimal overall impact on students' mathematical performance, highlighting the persistent challenge of competency in mathematics for this group. In contrast, language-responsive instruction, which integrated mathematics content with linguistic scaffolding, proved more effective, reducing cognitive load and anxiety while enabling students to demonstrate their abilities and communicate their thinking better.

The findings suggest that successful language-responsive mathematics teaching requires both strategic planning and flexible adaptation during instruction. Planning ensures that lessons align with students' linguistic and mathematical needs, while the intervention itself provides a practical framework for translating these plans into meaningful classroom interactions. This reinforces the argument that language-responsive teaching should be an integrated component of mathematics instruction, rather than an isolated add-on. However, the dual demands of processing mathematical concepts and language within a CLIL framework posed significant challenges, particularly for

AEN learners operating in a second language. These findings emphasize the necessity of carefully planned integration of content and language, supported by structured scaffolding, to meet the specific needs of AEN learners effectively (Ní Ríordáin et al., 2024).

Moreover, the findings underscore the importance of expanding and leveraging learners' plurilingual repertoires to deepen mathematical understanding and communication, and to foster bilingual disciplinary literacies (Coyle & Meyer, 2021). For AEN learners in IME, such approaches are particularly urgent as they navigate the dual challenges of linguistic and cognitive demands. This necessitates a deeper exploration of disciplinary literacies, which not only support the acquisition of specialized terminology, but also develop the broader communicative competencies essential for meaningful engagement with subject-specific content, particularly in bilingual and CLIL contexts (Llinares et al., 2012; Hüttner & Dalton-Puffer, 2024).

These competencies comprise the skills, knowledge, and abilities needed to understand and produce subject-specific language (Moje, 2015). We draw on Evnitskaya and Dalton-Puffer's (2023) definition of subject literacy, which they comprehend as, "the interplay between thought processes or thinking skills on the one hand and their inter-subjectively accessible expression (chiefly via language) on the other" (p. 311). Acquiring disciplinary knowledge for AEN learners is a challenge, and this challenge is compounded in CLIL settings. A core question arises in relation to how we support AEN mathematics learners in acquiring disciplinary knowledge and an ability to communicate this knowledge in potentially more than one language.

An operational framework for planning for AEN mathematics learners in CLIL contexts

The proposed framework is designed as a model to assist teachers working in CLIL contexts to plan the teaching and learning of language and content requirements for mathematics learners in their context, particularly those who work with students who may have additional needs related to language or, especially, literacies. The following framework encompasses several theoretical principles which work together to support the planning of teaching and learning mathematics in CLIL contexts. This approach is characterized as one that is language-responsive (Erath et al., 2021). Rather than focusing on the explicit pre-teaching of language, this approach encourages teachers to consider the ways in which learners need to use language in disciplinary-appropriate ways to participate in mathematical practices (Moschkovich, 2015). Here, language is the currency

of mathematical reasoning and communication, facilitating learners' gradual socialization into the discipline.

Mathematical literacy, as conceptualized here, encompasses three intertwined components: mathematical proficiency, mathematical practices, and mathematical discourse (Moschkovich, 2015). This expanded perspective moves beyond a narrow focus on vocabulary or procedural fluency, instead emphasizing learners' ability to engage in mathematical reasoning, represent ideas across multiple modalities, and participate in discipline-specific discourse. This article provides a structured framework for the planning of communicative competence in CLIL/immersion settings as part of Moschkovich's (2015) mathematical discourse. Our rationale for this specific focus is based on the premise that learning is inextricably linked to communication – without the ability to express their understanding, learners cannot fully demonstrate knowledge, participate in mathematical practices, or become socialized into the discipline. Our framework provides practical guidance for developing mathematical discourse in CLIL/ immersion settings, with broader applicability to non-immersion contexts. Furthermore, it extends Moschkovich's model by addressing how these practices can be adapted for learners with AEN, particularly those with cognitive or language-related needs.

The model is theoretically grounded in three approaches to the teaching and learning of language, communication, and literacies, which we examine below within the context of planning for mathematical literacy. The first theoretical approach is communicative competence in educational settings. This entails knowing how to receive and process meaning (reception) in communicated written or spoken language; to mediate (mediation) that written and spoken language in order to make sense of it; to communicate new meanings (production) through speaking and writing in ways that are appropriate to the task, discipline, age and stage of learners and to maintain communication despite linguistic limitations (Richards, 2006). In CLIL settings, planning for language development in this way helps develop learners' ability to engage in mathematical practices such as explaining, modeling, and reasoning, all of which require communication.

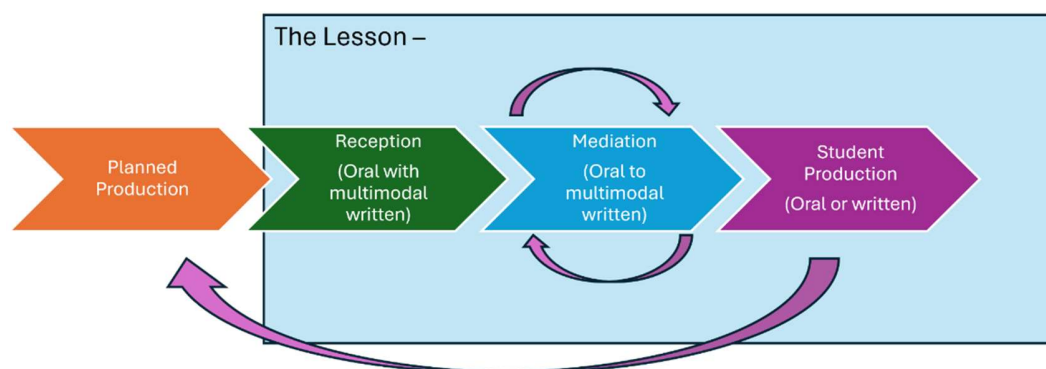
To develop communicative competence in CLIL settings, the CAPA (contextualization, awareness, practice, and autonomy) model, as proposed by Tedick and Lyster (2019), offers educators a way to enact the development of communicative competence. Combining CAPA principles with communicative competence in the model highlights the need for an initial planning

stage that integrates these elements. This recommendation stems from the case study findings, where scaffolding was most effective when cognitive, linguistic, and representational demands were aligned in pre-lesson planning (Ní Ríordáin et al., 2024).

For CAPA to be successfully implemented in CLIL contexts, the context of the multilingual setting in which it is enacted must be considered. Pluriliteracies for Teaching and Learning (PTL) (Meyer et al., 2015) recognizes that, in CLIL settings, all learners develop subject-specific literacies *across* languages, not just in the language of instruction. In CLIL mathematics classrooms, learners will undoubtedly develop conceptual understanding, strategic competence, and adaptive reasoning multilingually. While the ultimate communication of their learning might be in the CLIL language, the mediation of learning may take place in multiple languages, fostering deeper learning. Moreover, the sociocultural framework underpinning PTL aligns closely with Moschkovich's (2015) perspective, which also recognizes that mathematics instruction also encompasses other multimodal codes such as diagrams, gestures, and everyday language, all of which form part of learners' pluriliteracies. Together, these strategies promote the cognitive and discursive demands of mathematics while acknowledging learners' hybrid language practices.

Figure 1

The key stages of planning, teaching & learning for AEN mathematics learners in CLIL contexts



An additional contribution that this article makes to Moschkovich's perspective is the cognitive aspect of learning. In planning for AEN learners, it is essential to be both cognitively and linguistically responsive, designing tasks that encourage reasoning and conceptual understanding and strike a balance between content, language, and cognitive load. Moschkovich (2021) emphasizes that mathematical tasks should foster learners' participation in academic mathematical

practices, using discourse as a tool for learning rather than a barrier. By connecting mathematical literacy with these theoretical approaches, this framework seeks to create inclusive, cognitively engaging learning environments. The following subsections outline the stages of planning, teaching, and learning (Figure 1), grounded in these theoretical principles and tailored to the needs of AEN learners in CLIL mathematics classrooms.

Stage 1: Plan production of content and language

The planned production phase relates to the educator's readiness to meet the linguistic and content requirements of the students, including foreseeing potential difficulties and organizing lessons systematically. In CLIL contexts, this entails an added dimension of linguistic expertise for content instruction, as Morton (2024) elucidates, integrating general academic proficiency with specialized scaffolding methodologies. Teachers identify learners' existing language skills and their needs for real-world communication, particularly for expressing mathematical knowledge, establishing both content and language objectives.

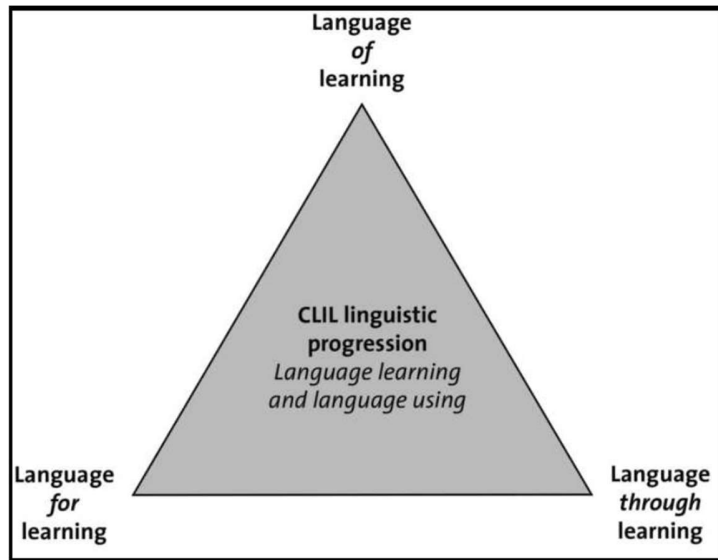
At the pre-CAPA planning stage, content teachers must integrate their pedagogical content knowledge with considerations of the mathematical genre's language demands, focusing on both the students' required language and the teacher's language to facilitate, advance, and evaluate learning. Utilizing Moschkovich's (2015, 2021) sociocultural framework, mathematical literacy is conceptualized as an integrated process involving mathematical proficiency, mathematical practices, and mathematical discourse. This view expands beyond procedural fluency, emphasizing reasoning, representation, and discourse as essential components of mathematical learning.

The Language Triptych (Coyle et al., 2010, p. 36; see Figure 2) is useful in helping teachers strategically to plan for and sequence learning objectives and support the conceptualized languaging (i.e., using language in specified contexts) for knowledge construction (Dalton-Puffer, 2007). It is composed of three language uses – the language *of* learning (needed to understand and communicate content), the language *for* learning (used to interact in the classroom), and the language *through* learning (new language provided to address learners' communication gaps) – which can be useful for structuring teacher planning. These languages need to be considered in tandem with planning for mathematics, which requires explicit consideration of the mathematical

register, inclusive of precise terminology, multimodal representations (graphs, equations, and gestures), and structured discourse (Moschkovich, 2015).

Figure 2

Language Triptych (Coyle et al., 2010, p. 36)



Planning must also account for language use within the broader context of schooling languages. To develop proficiency in the language of schooling, learners must gain mastery over both general academic and subject-specific language, transitioning dynamically between these forms (Prediger & Wessel, 2013). Colloquial language plays a vital pedagogical role, especially in mathematics, by enabling learners to express content and negotiate meaning in accessible and concrete ways. Planning for CLIL teaching should incorporate colloquial language as a starting point or strategy to demystify complex content and connect it to learners' everyday experiences, particularly benefiting AEN students.

Planning must also account for mathematical literacy as a discipline-specific form of academic literacy. Moschkovich (2015) emphasizes that students need opportunities to engage with both the macro-genres of mathematical tasks (e.g., problem solving, reasoning) and the micro-genres embedded within them (e.g., categorization, explanation, comparison). Dalton-Puffer (2016) has proposed a construct of Cognitive Discourse Functions (CDFs) that consists of seven general types of thought processes (categorize, define, describe, evaluate, explain, explore, and report), all of

which underpin various subject curricula and teaching and learning activities in classrooms. (Dalton-Puffer et al., 2018).

CDFs, as Coyle and Meyer (2021) argue, function as micro-genres within this framework. Macro-genres, such as problem solving, are complex text types built from smaller micro-genres (Coffin & Donohue, 2014). Learners must therefore master multiple micro-genres effectively to communicate and succeed in mathematics' macro-genres. CDFs, such as categorizing, defining, and explaining, can serve as critical tools for planning for scaffolding interactions. These functions allow learners to engage cognitively while negotiating the language required for mathematical reasoning. Teachers can adapt the language used to express CDFs to suit students' language proficiency, simplifying or contextualizing it where necessary.

Adopting such a genre-based planning approach for mathematical literacy begins with teachers modeling a typical response expected of students at a given age and stage who have mastered the content and language. Teachers then analyze their output to identify key discourse, sentence, and lexicogrammatical features aligned with subject-specific expectations. Discourse features include functions such as narration, explanation, and description, while sentence features involve structural patterns like linking words ("and", "also") or sequencing terms ("first", "second", "third"). Lexicogrammatical aspects, encompassing vocabulary and grammar (nouns, verbs, adjectives), work together to create meaning. Below is an example of a teacher's articulation on learners' required conceptual knowledge of fractions and their relationships, followed by a genre-based analysis of its features.

Initial articulation by teacher

Three-quarters is a fraction that represents three out of four equal parts of a whole. The numerator (three) indicates the number of parts being considered, while the denominator (four) represents the total number of equal parts that make up the whole. This fraction is greater than one-half, but less than one whole.

Discourse features

Explanatory text where the phenomenon of fractions is explained and broken down into its constituent parts and then applied to an example.

Sentence features

Simple declarative sentences are used to break the concept down into manageable parts.

Lexicogrammatical features

- a. A fraction represents [parts] out of [total parts] into which the whole is equally divided. (Relational structure)
- b. The number on the top (numerator) indicates the selected parts; the number on the bottom (denominator) indicates the total number of equal parts. (Partitive structure)
- c. Three-quarters is greater than one-half. (Comparative structure)
- d. [Fraction] is more than [Fraction], but less than a whole. (Comparative structure)

After completing this analysis, teachers can consider how CDFs feature in the discourse. CDFs are valuable for planning language in CLIL contexts as they allow students to engage with subject-specific facts, concepts, and language simultaneously (Dalton-Puffer, 2013). CDFs enable teachers to scaffold content based on learners' linguistic and cognitive readiness (Coyle & Meyer, 2021, pp. 81–83, see Table 1 for examples). For instance:

- Simplifying “Three-quarters is a fraction that represents ...” to “Three-quarters means 3 out of 4 equal parts”.
- Using real-world analogies, such as dividing a chocolate bar, to compare fractions.

Table 1*Lexicogrammatical features and their relationship to CDFs*

Lexicogrammatical feature	CDFs	Linguistically differentiation function
a. A fraction represents [parts] out of [total parts] into which the whole is equally divided. (Relational structure)	Classifying	A fraction describes how many parts we have relative to the total number of equal parts that form a whole.
b. The number on the top (numerator) indicates the selected parts; the number on the bottom (denominator) indicates the total number of equal parts. (Partitive structure)	Explaining	The denominator tells us how many equal parts the whole has been divided into, while the numerator tells us how many of these parts we have.
c. Three-quarters is greater than one-half. (Comparative structure)	Comparing and Contrasting	Imagine you have one chocolate bar divided into four equal pieces and another chocolate bar of the same size divided into two equal pieces. If you take three pieces

		from the first bar and one piece from the second, which is larger?
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Mathematical literacy integrates multiple modalities – gestures, drawings, equations, and language (Moschkovich, 2015). For AEN learners, scaffolding multimodal connections is critical. Teachers should incorporate tasks that allow students to:

- use visual aids to represent e.g., fractions (pie charts, number lines);
- transition between everyday language (“dividing a chocolate bar”) and mathematical terms (“equal parts”);
- participate in discourse practices like justifying their answers (“Three-quarters is more than one-half because ...”).

This multimodal approach ensures that mathematical concepts are both linguistically accessible and cognitively rigorous (Moschkovich, 2021). Once these content and language expectations have been established, the teachers’ planning can be enhanced by considering the CLIL learning through a pluriliteracies lens. For example, when using comparative structures, teachers might connect the target language with learners’ L1, highlighting similarities and differences in grammar and expression. Below is an example of both structures in English (A) and Irish (B). Here, both the English and Irish structures are similar in that they contain a copular verb (‘tá’ and ‘is’), followed by a comparative marker (more/níos mó) and a preposition (than/ná). However, two key differences are that the copular ‘is’ in English signifies the 3rd person singular, whereas the Irish ‘Tá’ has no singular, plural or person marking. The second key difference is that the copular verb in Irish appears at the beginning of the sentence before the subject (verb + subject + complement), whereas in English it appears in the second position (subject + verb + complement).

- A. [Fraction] is more than [Fraction]
 B. Tá [Codán] níos mó ná [Codán]

While these grammatical details may not be explicitly taught, highlighting them helps reinforce connections between learners’ linguistic repertoires and the mathematical content. Overall, by incorporating CDFs, multimodal resources, and genre-based planning, this stage prepares teachers

to create inclusive, linguistically responsive mathematics lessons that engage all learners, particularly those with additional educational needs (Ní Ríordáin et al., 2024).

Stage 2: Oral reception → structured oral production and interaction (contextualization and awareness)

The second stage, contextualization, involves delivering content and language through planned teacher talk in a communicative context (Van Patten, 2017). Teachers model content and language, connecting with prior knowledge and real-world experiences, creating a foundation for mathematical literacy, while learners engage in scaffolded activities that encourage interaction with new material. Utilizing Moschkovich's (2015, 2021) framework, this stage integrates mathematical proficiency, mathematical practices, and mathematical discourse to ensure learners actively engage with mathematical concepts, representations, and reasoning.

Contextualization introduces new content and language by linking it to learners' prior knowledge. For fractions, this might involve concrete, familiar examples such as slicing a pizza or a cake to explore part-whole relationships. These examples use everyday language to connect abstract mathematical concepts to learners' lived experiences. As Moschkovich (2021) notes, using familiar contexts and colloquial language enables learners to bridge the gap between informal understanding and academic mathematical discourse.

Subject-specific language identified during planning is then introduced gradually, linking it to concrete concepts before moving to more abstract ideas. Differentiated use of CDFs allows learners to start with familiar linguistic structures, progressing to more academic representations. Teachers should model the required language and limit input initially to ensure learners can see and hear these structures used meaningfully and implicitly outline the content and language expectations for learners in order to begin raising their awareness. This progression aligns with Moschkovich's (2015) emphasis on using multiple modes of representation – visuals, gestures, and verbal explanations – to support conceptual understanding and discourse development.

After providing language and content input, it is crucial to include an often-overlooked oral practice phase to develop communicative competence. Oral practice is a vital component for fostering mathematical discourse, as it allows learners to explore concepts and language. This phase allows learners to “play” with new language through structured repetition, both with teachers

and peers, in planned interaction tasks (Saunders & O'Brien, 2006). Interaction, apart from developing their communicative competence, provides natural feedback as peers respond in either L1 or L2, helping learners refine their language and assess their communication success (Sharple, 2021). For example, in a fractions lesson, learners might describe objects divided into parts (e.g., food, blocks), narrating their observations using language structures like, "[Fraction] represents [parts] of a whole". Interaction with peers and teachers provides opportunities for AEN learners to refine their discourse through natural feedback and repetition. Moschkovich (2015, 2021) highlights that mathematical literacy is inherently multimodal, involving spoken language, visual representations, and symbolic systems. To support this, teachers should design activities where learners collaboratively engage with multiple modalities. For instance, learners could label parts of a pie chart while explaining the fractions represented, combining visual and verbal modes to solidify their understanding. Teachers should also differentiate these tasks to meet the diverse needs of AEN learners, ensuring that all students can participate meaningfully.

After initial practice, the teacher highlights key discourse, sentence-level, or lexicogrammatical features, an essential part of the CAPA model, where explicit reference is made to key linguistic forms in context. According to Moschkovich (2015), mathematical discourse involves precise and situated meanings that emerge from participation in mathematical practices. It is this explicit 'noticing' of form that helps strike a balance between the teaching of content and language (Swan, 2011), and helps learners analyze and reflect on its role in discussing the concept. Teachers should guide learners to analyze how specific language forms, such as comparative structures or relational terms, are used to convey mathematical ideas.

In bilingual contexts, referencing learners' L1s enhances their ability to negotiate meanings in both languages, fostering deeper mathematical literacies. As Moschkovich (2021) explains, mathematical proficiency is not limited to the dominant language; hybrid language practices, including translanguaging, can enrich learners' engagement with content. Interaction tasks naturally provide feedback as peers respond, either in L1 or L2, refining learners' communication and comprehension. Teachers should use this feedback to evaluate both content and language development, ensuring that learners are progressing in their ability to articulate mathematical ideas. By embedding mathematical literacies into contextualization and oral interaction, this stage ensures that learners not only acquire mathematical language, but also develop the proficiency,

practices, and discourse needed to participate in mathematical learning. This integrated approach aligns with Moschkovich's (2015) emphasis on multimodal and socially situated mathematical literacy, creating a supportive environment for AEN learners in CLIL contexts.

Stage 3: Mediation between oral reception, oral production and written production (practice)

The third stage, awareness and practice, focuses on mediation, a process where learners cycle through reading, discussing, and writing while engaging with mathematical concepts and language. This stage emphasizes developing mathematical literacies (Moschkovich, 2015, 2021) with learners actively participating in explaining, reasoning, and representing mathematical ideas across different modes of communication. Through scaffolded mediation, learners' transition between oral and written modes of expression, gradually developing their ability to use formal mathematical language.

Mediation actively engages learners in collaborative and cognitively stimulating activities, supporting language comprehension and content mastery provided appropriate integrated content and language scaffolding is provided. This could include the use of sentence starters that include CDFs written at a level that aligns to their current level of mathematical literacies. For younger learners, this process helps develop an academic register, the precise, subject-specific language necessary for mathematical reasoning (Moschkovich, 2015). Gibbons (2002) notes that shifting from informal oral forms to formal written ones is crucial for building subject-specific literacies.

For example, when working with fractions, learners might begin by orally describing multimodal representations (e.g., elements of a pie chart) and then transition to writing structured explanations using scaffolds like cloze activities (e.g., see Table 2). Other activities might include pairs or small groups of learners explaining their reasoning aloud while solving problems, using prompts such as "How do you know that three-quarters is more than one-half?" or "Why does the bottom number represent the total number of parts?". Teachers can provide sentence starters or frame these discussions around specific CDFs (e.g., explaining or justifying).

Similarly, learners can be encouraged to link mathematical ideas across modalities, such as writing about fractions represented visually in diagrams. For instance, they might label parts of a fraction model (e.g., "This part is one-fourth") or write a description of the model (e.g., "One-fourth means one part out of four equal parts"). This transition reinforces connections between the mathematical

concepts and their linguistic representations, supporting conceptual understanding and fluency in the mathematical register.

Table 2

Example of cloze type activity in English and Irish

Sentence Completion Task	Partial Word Bank	Expanded Word Bank
A fraction is a way to represent a part of a whole: A fraction represents _____ parts out of _____ equal parts into which a whole has been divided. <i>Léiríonn codán _____ chuid as _____ chuid chothroma ina bhfuil an t-iomlán roinnte.</i>	three, four <i>trí, ceathair</i>	one, two, three, four, five, six, seven, eight, nine, ten <i>aon, dó, trí, ceathair, cúig, sé, seacht, ocht, naoi, deich</i>
Numerator and Denominator Identification: The number on the top, called the _____, tells us how many parts we have. The number on the bottom, called the _____, tells us how many equal parts the whole has been divided into. <i>Taispeánann an _____, an uimhir ar barr, cé mhéad cuid atá againn. Taispeánann an _____, an uimhir ar bun, cé mhéad cuid chothroma ina bhfuil an t-iomlán roinnte.</i>	numerator, denominator <i>uimhreoir, ainmneoir</i>	numerator, denominator, fraction, parts, total <i>uimhreoir, ainmneoir, codán, codanna, iomlán</i>
Comparing Fractions: _____ is greater than _____ because it represents more parts of the whole. <i>Tá _____ níos mó ná _____ mar go léiríonn sé níos mó codanna den iomlán.</i>	one-half, one-quarter <i>leath, ceathrú amháin</i>	one-half, one-third, two-thirds, three-quarters <i>leath, trian amháin, dhá thrían, trí cheathrú</i>
Independent Construction: A fraction is a way to express _____. <i>Is bealach é codán chun _____ a chur in iúl.</i>	(Learner writes their own response)	(Open-ended)

This encourages independent use of subject-specific language while engaging with CDFs like classifying and explaining, integrating content and language learning. Essential of this stage is helping learners notice and reflect on key discourse, sentence-level, and lexicogrammatical features within the mathematical tasks. Moschkovich (2021) emphasizes that this analysis should

remain situated in meaningful mathematical activity. Teachers might explicitly model how features support mathematical reasoning and ask learners to apply them in their explanations or written work. By integrating mathematical literacies into this stage, awareness and practice tasks simultaneously support conceptual understanding and language development. Learners develop their ability to participate in mathematical discourse, connect representations, and articulate reasoning, laying the groundwork for further engagement with mathematics.

Stage 4: Written and oral production (autonomy)

Finally, the autonomy stage sees learners demonstrate mastery of both mathematical content and communication, with teacher assessments addressing any remaining gaps, ensuring alignment with learning outcomes and preparation for real-world application. This phase emphasizes the integration of mathematical literacies as learners produce written and oral outputs that reflect their conceptual understanding and ability to communicate within the conventions of the discipline (Moschkovich, 2015).

In the final lesson phase, teachers assess learning both formatively and summatively, considering content and language aims. Learners should complete a task demonstrating both, with scaffolding adapted to their abilities. Assessment should evaluate their mathematical proficiency, mathematical practices, and their ability to communicate it using the genre expectations (mathematical discourse) set during planning. For example, when assessing a task involving fractions, teachers could examine not only the accuracy of calculations, but also how learners articulate their reasoning. Mathematical literacy requires learners to integrate multiple modes of communication – oral, written, visual, and symbolic – and it is important that these modes and multiple modes are integrated into assessments (Moschkovich, 2015). For example, a learner might be asked to explain how they solved a fractions problem using a diagram. Teachers assess not only the correctness of the solution, but also how well the learner integrates verbal explanations with visual or symbolic elements.

As Moschkovich (2021) highlights, learners often begin by using colloquial or everyday language to express mathematical ideas, transitioning to more formal, academic registers over time. If learners rely on familiar, colloquial language instead of subject-specific terms, this should be addressed in future lessons with additional scaffolding. Mastering subject-specific genres is

crucial, as it provides learners access to “genres of power” (Dalton-Puffer, 2013, p. 233) essential for social, political, and cultural opportunities.

Conclusion

This article has addressed the challenges and gaps in planning for AEN mathematics learners within CLIL contexts. It has emphasized the dual demands of content and language learning, which require targeted instructional strategies to support equity and inclusion (Pérez Cañado, 2023). The proposed operational framework integrates theoretical and communicative approaches – communicative competence (Richards, 2006), CAPA (Tedick & Lyster, 2019), and pluriliteracies (Meyer et al., 2015) with mathematical literacy (Moschkovich, 2015) – to align language and content learning with the cognitive needs of AEN learners. The transition from theoretical planning to practical implementation in mathematics education presents several challenges, including the need for flexibility, teacher autonomy, and contextual adaptation.

While our framework serves as a structured guide for lesson planning and implementation in mathematics classrooms, its application in real-life teaching scenarios requires empirical validation. To bridge the current knowledge gap, future research should examine this framework in practice, expand on its application to other multilingual settings and pedagogies, and explore its long-term impact on learner outcomes and achievements. By addressing the articulation and intersection of language, cognition and content in mathematics, it is intended that this approach more generally contributes to creating more inclusive and effective CLIL environments (Ní Riórdáin et al., 2024).

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